

Project Executable Files

Date	2 July 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted(Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes (SQLite Database)
5	Environment configuration file or (.env.example similar)	Yes
6	Deployed application URL (if hosted)	No (Local Deployment)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure :

SMART LENDAR/

```

|
|— .venv/
|— Dataset/
|   |— loan_approval.csv
|— eda_plots/
|— flask/
|   |— static\css/
|   |— templates/
|   |— .env
|— app1.py
|— loan_model.pkl
|— .gitignore
|— edaanalysis.py
|— loan_model.pkl
|— loans.db
|— modelbuilding.py
|— preprocess.py
|— test.py
|— train_model.py
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Local Flask Application (http://127.0.0.1:5000)
Login Credentials (Demo)	Not Applicable (No Login Required)
Platform/Hosting Provider	Local Flask Development Server

Repository Link	https://github.com/ReddimalliRuthika/Smart-Lender
Demo Video Link	https://drive.google.com/file/d/1on9a8P107aACVmzxFE922P44lORHm50H/view?usp=sharing

Step 4: Run Instructions

Pre-requisites

- Python 3.14 or above
- Visual Studio Code
- Flask
- XGBoost
- SQLite
- Git (Optional)

Installation Steps:

Step 1

Clone the repository.

```
git clone <GitHub Repository URL>
```

Step 2

Open the project folder.

```
cd Smart-Lender
```

Step 3

Create a virtual environment.

```
python -m venv venv
```

Step 4

Activate the virtual environment.

Windows

```
venv\Scripts\activate
```

Step 5

Install all dependencies.

```
pip install -r requirements.txt
```

Step 6

Create a .env file and configure:

```
EMAIL_ADDRESS=your_email@gmail.com
```

```
EMAIL_PASSWORD=your_app_password
```

```
MODEL_PATH=saved_model/xgboost_model.pkl
```

```
FLASK_SECRET_KEY=your_secret_key
```

Step 7

Run the application.

```
python app.py
```

Step 8

Open the application in your browser.

```
http://127.0.0.1:5000
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Model prediction depends on training dataset quality	Can be improved by training with a larger and more diverse dataset
2	Application runs locally by default	Can be deployed on IBM Cloud, Render, or another cloud platform
3	Email notifications require a valid Gmail App Password	Configure a valid App Password in the .env file